

6	(a)	(i)	$\frac{V_S}{V_P} = \frac{N_S}{N_P} \rightarrow V_{S,peak} = 170 \left(\frac{3500}{2000} \right) = 297.5 \text{ V}$ $V_{S,rms} = \frac{297.5}{\sqrt{2}} = 210 \text{ V}$	M1 A1
		(ii)	$V_{S,rms} = I_{S,rms} R \rightarrow I_{S,rms} = \frac{210}{130} = 1.62 \text{ A}$ mean power output = $\langle P \rangle_{output} = V_{S,rms} I_{S,rms} = (210)(1.62) = 340 \text{ W}$ Power input = Power output (ideal transformer) peak power input = $2 \times \langle P \rangle_{output}$ $V_{P,peak} I_{P,peak} = 2 \times 340 \Rightarrow I_{P,peak} = \frac{(2 \times 340)}{170} = 4.00 \text{ A}$	M1 A1
	(b)	As the number of turns in the secondary coil is halved while input voltage and number of turns in the primary coil remain the same, the induced voltage in the secondary coil is halved. The power output is proportional to the square of the voltage. The resistance of the heater is unchanged, so the new mean power dissipated is $\frac{1}{4} P$.		B1 A1
			Total:	7

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7	(a)	When the energy level of atom or system (not electron) at $(n = \infty)$ is 0 eV/ this is where	B1
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